

Cyber Security Internship – Task 1

Local Network Port Scanning using Nmap

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Objective:

To scan my local network and find open ports on connected devices. Learn how attackers or security professionals identify exposed services using tools like Nmap.

Step-by-Step Log:

1. First, I installed Nmap on my Kali Linux system.

Command used:

```
sudo apt install nmap
```

2. Found my local IP range using:

```
ip addr show
```

My IP was: 192.168.11.144

So I assumed subnet: 192.168.1.0/24

3. Ran the TCP SYN scan:

Command:

```
nmap -sS 192.168.1.0/24
```

4. The scan took around 20 seconds. Got results showing some active devices on the network:

- 192.168.1.7 (My machine)

5. Observed open ports on router and PC:

- 52/tcp

6. I noted these ports and researched what services they usually represent.

7. Optional:

I opened Wireshark and captured packets while running Nmap. I could actually see SYN packets going out and SYN-ACK coming back from some IPs. That was interesting.

8. Saved my scan output using:

```
nmap -sS 192.168.1.0/24 -oN scan-result.txt (example like this your ip should be different)
```

Learnings:

- Understood how TCP SYN scanning works.
- Realized that many devices have open ports by default.
- Some services like NetBIOS and SMB could be risky if exposed.
- Gained practical experience with Nmap.
- Wireshark helped me visually see what a scan looks like at the packet level.

Conclusion:

This task helped me understand real-world scanning techniques and how important it is to monitor and restrict unnecessary services. I'll now explore how to block or secure these open ports using a firewall.